



HermiteFlow

Product & Launch Guide — everything you and your clients need to know about what we're building, how it works, and what's required to ship it.

Prepared for	HermiteFlow stakeholders & clients
Product	HermiteFlow — online invoicing SaaS
Stage	Marketing site live; building the product
Stack	TypeScript · React · Supabase · Polar.sh · Stripe · Resend · Vercel
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Confidential — for internal and client use. Companion technical docs: [PRODUCT_PLAN.md](#), [SETUP_GUIDE.md](#), [API.md](#).

1. Executive summary

HermiteFlow is a fast, focused online invoicing platform for freelancers, agencies, and small finance teams. Users create polished, branded invoices, send them by email with a built-in payment link, and get paid faster through automated reminders — all from a clean, Linear-grade dashboard.

The public marketing website is already built and deployable. This guide covers building the actual product: a secure, multi-tenant SaaS with accounts, subscription billing, online payments, automation, and a public API — engineered with modern, scalable best practices in TypeScript.

What "done" looks like (v1): a customer signs up, creates their organization, adds clients, issues invoices, gets paid online, and is billed monthly for their plan — with email automation and a documented API, on infrastructure that scales automatically.

Who this is for

- You (owner): understand scope, timeline, costs, and exactly what to set up.
- Your clients / customers: understand what the product does, how data is protected, and how billing & payments work.

2. What the product does

Capability	Description
Accounts & teams	Sign up, create an organization, invite teammates with roles (owner, admin, member, viewer).
Clients	Manage customers, contacts, addresses, and payment terms with full invoice history.
Invoices	Branded invoices with line items, automatic VAT/tax, multi-currency totals, and PDF export.
Send & get paid	Email invoices with a secure online payment link; clients pay by card/bank; status flips to paid.
Automation	Scheduled reminders before/after due dates; automatic overdue flagging.
Dashboard	Live revenue, outstanding balances, and overdue counts at a glance.
Subscriptions	Self-serve plans (Starter free, Pro, Business) with upgrade/downgrade and tax handled for you.
Public API	Programmatic access to clients, invoices, and payments for integrations.

3. How it works (architecture)

The product is built as a single, typed system with one shared business-logic core serving three entry points: the app UI, a public API, and payment webhooks. It runs on serverless infrastructure that scales automatically with demand.

Layer	Technology	Role
Web app	React + Vite + Tailwind	The dashboard & marketing site (already built).
API	tRPC + REST (TypeScript)	Typed first-party API + public REST API for integrations.

Database	Supabase Postgres	All data, isolated per organization with row-level security.
Authentication	Supabase Auth	Email, magic link, and Google sign-in with secure tokens.
Subscriptions	Polar.sh	Bills customers for their plan; handles sales tax/VAT as merchant of record.
Invoice payments	Stripe	Lets your customers' clients pay invoices online by card/bank.
Email	Resend	Invoices, receipts, reminders, and team invites.
Files	Supabase Storage	Stores generated invoice PDFs securely.
Hosting & jobs	Vercel + Vercel Cron	Global serverless hosting and scheduled automation.

Engineering principles

- One service layer, three doors. App, API, and webhooks share the same validated logic — no duplication.
- Multi-tenant & isolated. Every record belongs to an organization; the database guarantees isolation.
- Stateless & serverless. Scales horizontally and automatically; no servers to babysit.
- Typed end to end. TypeScript + schema validation catch errors before production.

4. Security & data protection

- Per-tenant isolation enforced at the database level (row-level security).
- Secure auth via Supabase (industry-standard JWTs); secrets never in the browser.
- API keys stored hashed, shown once, scoped, and revocable.
- Verified, idempotent webhooks for all payment events.
- Input validation on every request; rate limiting on sensitive routes.
- Encrypted in transit and at rest; secrets managed in Vercel/Supabase, not in code.
- A security review is run before every release.

5. What you need to set up & connect

To go live, a few third-party accounts must be created and connected. Full click-by-click steps are in docs/SETUP_GUIDE.md; this is the summary and the order to do them in.

Service	Purpose	What you provide	Order
Supabase	Database, auth, file storage	Project URL + keys + DB connection	1st
Resend	Sending email	API key + verified sending domain	2nd
Polar.sh	Subscription billing	3 products + access token	3rd
Stripe	Invoice payments	API keys + webhook secret	4th
Vercel	Hosting + automation	Import repo + env vars + domain	5th
Domain	Web + email address	Domain + DNS to Vercel/Resend	6th
Sentry (optional)	Error monitoring	Project DSN	7th

Do Supabase first. As soon as the Supabase project exists and its keys are shared, engineering begins Phase 1. The remaining accounts can be set up in parallel.

6. Build roadmap & timeline

We execute linearly and ship as we go — each phase is independently deployable, so progress is visible continuously rather than in one big-bang release.

Phase	Outcome	Needs from you
0 — Done	Marketing website + deployment config	—
1	Database + accounts + organizations (multi-tenant foundation)	Supabase keys
2	Core invoicing & clients live in the multi-tenant app	—
3	Subscription billing (Polar) + online invoice payments (Stripe)	Polar + Stripe keys
4	Public REST API + API keys + interactive docs	—
5	Email automation, reminders, PDF storage	Resend + domain
6 — Launch	Security review, testing, monitoring !' go live	Sentry (optional)
7	Recurring invoices, multi-currency, audit log, analytics	—

7. Running costs (estimate)

The stack is chosen to start near-free and scale with revenue. Typical early-stage monthly cost:

Service	Free tier	Paid (when you grow)
Supabase	Yes (hobby)	~\$25/mo Pro
Vercel	Yes (hobby)	~\$20/mo Pro
Resend	3,000 emails/mo	from ~\$20/mo
Polar.sh	No monthly fee	% of subscription revenue (merchant of record)
Stripe	No monthly fee	~2.9% + 30c per payment
Domain	—	~\$12/year
Sentry	Yes	from ~\$26/mo

Payment processors only charge when you make money. You can launch on free tiers and upgrade as usage grows.

8. Immediate next steps

- You: create the Supabase project (Setup Guide §1) and share the keys.
- You (parallel): start Resend, Polar, and Stripe accounts.
- Engineering: begin Phase 1 (database, auth, organizations) once Supabase is ready.
- Together: review each phase as it ships; adjust scope toward launch.

